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Enhancing Operational Efficiency in Hydraulic Fracturing: A Predictive Modeling Approach for Fracturing Design in the Sirikit Oil Field

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Abstract

This paper presents a predictive modeling approach aimed at enhancing operational efficiency in hydraulic fracturing design, specifically tailored for tight sand reservoirs in the Sirikit Oil Field, Thailand. The methodology integrates deterministic data from pre-fracture diagnostic tests, including Mini-Falloff Tests (MFTs) and Step-Rate Tests (SRTs), to develop empirical correlations between fluid efficiency and optimal pad percentage. A comprehensive dataset encompassing over 200 fracturing stages across 100 wells was analyzed, revealing a strong linear relationship between fluid efficiency from SRTs and pad volume requirements. To improve predictive accuracy and applicability, the dataset was stratified into geological compartments based on common depositional and structural characteristics. This compartmentalized modeling enabled the use of offset well data for design purposes, allowing the omission of calibration injection steps in appropriate scenarios.

Field validation in Well-A demonstrated the model's accuracy, with only a 2% deviation between predicted and actual pad percentages. Following this, the model was deployed across 15 wells and 26 stages in the N Area during 2023–2024, resulting in zero screen-out events, an average of 5–10 hours saved per stage, and a 7.6% reduction in fracturing costs. The results confirm that combining empirical modeling with geological segmentation can significantly improve treatment design precision, reduce operational risk, and enhance efficiency. This approach establishes a foundation for future advancements, including integration with multivariate regression, theoretical models, and machine learning techniques to support broader field-wide and cross-basin applications.

Background

Sirikit Oil Field's Hydraulic Fracturing Journey

The Sirikit Oil Field (S1), located in the S1 concession block in Kamphaeng Phet Province, is Thailand's largest onshore oil-producing field and a cornerstone of the country's domestic energy infrastructure. Discovered in 1981 and commencing production in 1983, the field is operated by PTTEP Siam Limited,

a subsidiary of PTT Exploration and Production (PTTEP). It has played a critical role in national energy security, currently contributing approximately 5% of Thailand's daily crude oil output. The field reached its peak production in 2013 and has since undergone various enhanced recovery initiatives, including hydraulic fracturing, to maintain output levels. The Sirikit field has produced over hundreds of million barrels of oil cumulatively and held remaining significant recoverable reserves, supported by ongoing technological and operational advancements aimed at maximizing recovery from both conventional and tight formations. Therefore, hydraulic fracturing technology was a crucial technology to maximize recovery from this area, especially tight formations.

Hydraulic Fracturing (HF) operations in the Sirikit Oil Field began in 2006. While early results were promising, subsequent campaigns in remote locations faced logistical hurdles, such as the need for mobile production facilities and unsuitable artificial lift methods. These challenges prompted a strategic shift in 2012 toward more accessible zones with existing infrastructure, supported by comprehensive geological and geomechanical studies. The 2016 campaign, achieving a four to five-fold increase in production and a gentler decline rate, with estimated ultimate recovery up to ten times higher than unstimulated wells.

Building on the success, a thorough review of wells led to refined selection criteria for future campaigns, balancing subsurface potential with structural integrity and development plans. A major milestone was achieved in 2018 with the development of a candidate selection guideline, which enabled the identification of optimal reservoirs for hydraulic fracturing. This tool was pivotal in the 2018/19 campaign, which delivered the highest production enhancement in PTTEP's history by strategically targeting multi-layered, low-permeability formations. This breakthrough marked the transition from campaign-based fracturing to routine operations, a practice that continues to this day.

Currently, tight sand reservoirs are found across various depths in the Sirikit Oil Field and represent approximately one-fifth of the field's total STOIP. Target reservoir selection for hydraulic fracturing is based on the following criteria: (1) expected pre-fracturing production of less than 50 barrels per day, (2) permeability below 10 millidarcies, and (3) porosity below 10%. At its peak, oil production from hydraulically fractured wells contributed up to 29% of the field's total output, while current production from these wells accounts for approximately 20% of total field production.

Fracturing Workflow in Sirikit Oil Field

Referring [Figure 1](#), the fracturing workflow in Sirikit Oilfield comprises various main steps. It begins with well data analysis (a), incorporating petrophysical logs, well location, and mechanical earth modeling to understand stress distributions. Using this input, a preliminary fracturing design (b) is created, including optimized fracture half-length, pump schedule, treatment pressure, and expected fracture geometry. Next, treatment execution (c) involves conducting diagnostic tests like mini fall-off, step rate, and calibration injections. These test results and computed parameters are then used in treatment evaluation (d) through pressure match and data analysis to validate and interpret reservoir response. Based on this, the plan may undergo treatment redesign with acquired parameters (e) to finally refine the design with actual response from the reservoir. Afterward, the main frac is executed (f) as per the optimized design. Finally, the main frac evaluation (g) ensures alignment between simulation and field results, completing a feedback loop that improves future fracturing strategies.

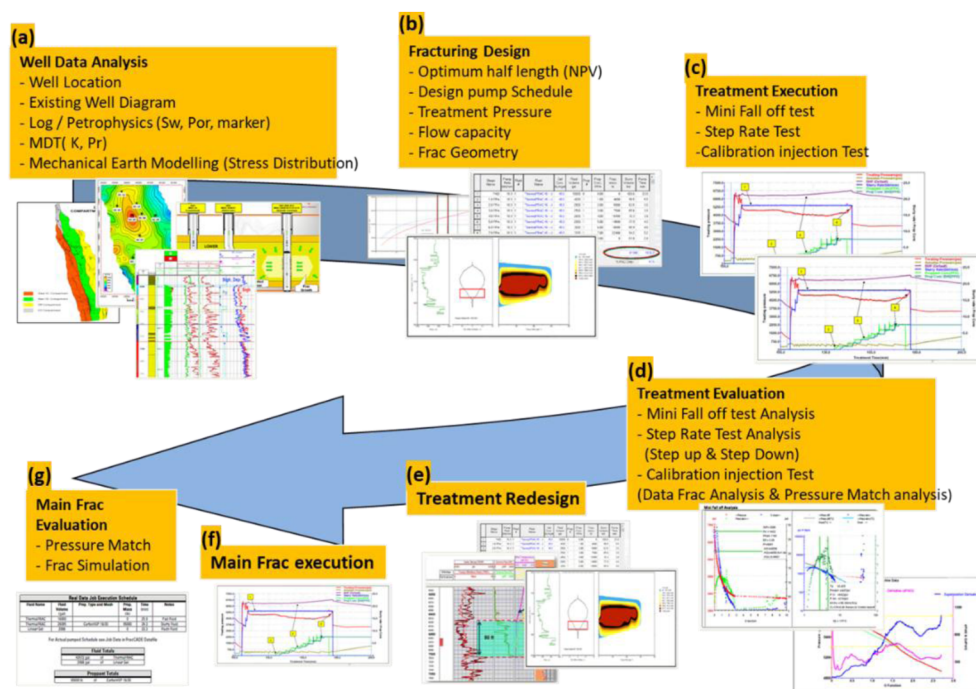


Figure 1—Fracturing Workflow in Sirikit Oil Field (a) Well Data Analysis; (b) Fracturing Design; (c) Treatment Execution; (d) Treatment Evaluation; (e) Treatment Redesign; (f) Main Frac Execution; (g) Main Frac Evaluation

Focusing Figure 1 (c), it highlights the data-deterministic tests that provide critical parameters for designing and executing the main hydraulic fracturing job (f). Referring Figure 2, the treatment execution sequence begins with a Mini-Falloff test, involving around 10 barrels of brine over 10 minutes to break the formation and observe early-time pressure behavior, helping identify formation breakdown pressure and closure tendencies. It is followed by a Step Rate Test, using approximately 100 barrels of brine over 30 minutes to determine the fracture initiation pressure and identify the onset of fracture propagation through rate vs. pressure relationships. The final and most comprehensive test is the Calibration Injection, which injects around 600 barrels of gel over 3–4 hours to simulate actual fracturing conditions. This step provides most of the key parameters, such as closure pressure, fluid efficiency, leakoff coefficient, and fracture geometry, that are essential for calibrating models and ensuring the main frac (step f) is executed with optimized treatment pressures, pump schedules, and proppant volumes. These carefully acquired parameters ensure the stimulation design is grounded in the reservoir's true behavior, reducing uncertainty and improving treatment effectiveness.

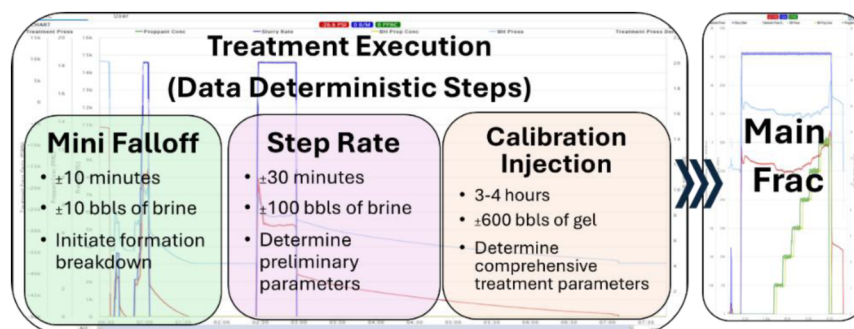


Figure 2—Treatment Execution Steps and Tests with details

Mini-Falloff Tests. A Mini-Falloff Test (MFT) is a diagnostic procedure commonly used in tight reservoirs, such as tight sandstones, to estimate key formation parameters using minimal fluid injection. It typically

involves injecting a small volume of water or brine at a controlled rate to initiate a short hydraulic fracture, followed by shutting in the well to monitor pressure decline over time. In the Sirikit Oil Field, MFTs are conducted before each fracturing stage using approximately 10 barrels of treated water. This quick, 10-minute process serves to confirm that the formation has been successfully fractured prior to continuing with further operations. The resulting pressure falloff is analyzed using techniques such as G-function or pressure derivative plots. However, due to the small fluid volume relative to that of the main fracturing treatment, the derived falloff parameters are generally considered unreliable for detailed reservoir characterization.

Step-Rate Tests. A Step-Rate Test (SRT) is used to identify the formation's fracturing pressure by injecting fluid at a sequence of increasing constant rates. In practice, fluid is pumped at one rate until pressure stabilizes, then the rate is increased to the next "step," and so on. The stabilized bottom-hole pressure is recorded at each step. Plotting injection pressure versus rate typically shows two distinct slopes: one at low rates (pre-fracture) and one at higher rates (post-fracture). The intersection where the slope of the low-rate branch meets the high-rate branch indicates the onset of fracturing, the minimum fracture extension pressure (often expressed as a pressure gradient, psi/ft). Similar to Mini-Falloff Tests (MFTs), the resulting pressure falloff can be analyzed using G-function or derivative methods. At the Sirikit Oil Field, SRTs are performed before each fracturing stage using a larger fluid volume, approximately 100 barrels of treated water, compared to MFTs. This step is used to identify fracture extension pressure and evaluate near-wellbore friction. The test typically takes around 30 minutes, and due to the higher fluid volume, the derived parameters are generally considered more reliable than those obtained from MFTs.

Calibration Injection Tests. A Calibration Injection Tests is an injection–shut-in test performed without proppant, immediately before the main treatment. In a calibration injection test, fracturing fluid is injected to break down the formation and initiate a short fracture, then the well is shut in and pressure is allowed to fall off. The goal is to "calibrate" the actual reservoir response under near-stimulation conditions with preliminary parameters from well data. During the shut-in (falloff) period, the pressure decline is analyzed with diagnostic plots to identify when the induced fracture closes. The closure point on the falloff curve corresponds to the reservoir's minimum principal stress (closure pressure). By examining the slope of the pressure decline (especially on a G-function or pressure-derivative plot), one also infers the fluid leakoff coefficient and hence the reservoir permeability. In summary, a properly-analyzed test yields fracture closure pressure, reservoir pressure, leakoff coefficient (and permeability), and an effective fracture fluid efficiency. These test-derived parameters (closure stress, leakoff/permeability, etc.) are the inputs needed to design the full fracture treatment.

Parameters from Treatment Execution Tests. The diagnostic tests yield the following key parameters for fracturing design on every steps of tests:

- **Closure (minimum horizontal) stress or pressure**, which controls the fracture's ability to stay open.
- **Leakoff coefficient (formation permeability)**, derived from the pressure decline rate after shut-in.
- **Fluid Efficiencies**, derived from the G-function plot.
- **Pad Percentage**, derived from the equation, $\text{Pad Percentage} = \frac{1 - \text{Fluid Efficiency}}{1 + \text{Fluid Efficiency}}$
- **Reservoir (pore) pressure**, inferred from late-time falloff pressure

These parameters are used to calibrate hydraulic fracture models and simulators. For example, closure stress, leakoff behavior, fluid properties and calculated volume derived from the tests are integrated into fracture propagation models to simulate the expected fracture geometry and behavior in the reservoir. Over time, as data from over 100 wells and 200 fracturing stages have been systematically collected, relationships

among key treatment parameters have anticipated, enabling further optimization of hydraulic fracturing operations in the Sirikit Oil Field.

Methods and Process

Data models from databases

From over 100 wells and 200 fracturing stages, the database full of parameters was investigated to find the useful relationships between parameters for prediction. Leveraging this database, particularly data from Mini-Falloff Tests (MFTs) and Step-Rate Tests (SRTs), predictive models were developed to estimate treatment volumes and optimize fracturing design. Specifically, a subset of the database from the S-sands* in the N Area*, containing over 50 stages, was used to build a model for predicting fluid efficiency and optimal gel volumes. This effort supported planning for over 10 wells and 20 stages scheduled in the 2023–24 period.

*"S-sands" and "N Area" are anonymized designations used to maintain data confidentiality.

With thorough investigation, significant relationships have been found between fluid efficiency from Step-Rate Tests and Pad percentages. The empirical linear correlation shows the strong empirical linear correlation between the parameters with an R^2 of 0.70 (Figure 3b), indicating a statistically significant and reliable trend. The effect has been expected from the physical trend between of higher fluid efficiency equals to lower pad requirements. In contrast, fluid efficiency from Mini-Falloff Tests (Figure 3a) showed insignificant relationship between fluid efficiency and pad percentage, with an R^2 of only 0.26, suggesting a statistically insignificant and less reliable trend, compared to 0.70 with fluid efficiency from Step-Rate Tests. This discrepancy is likely due to the smaller fluid volume used in Mini-Falloff Tests (Figure 2), which contacts only a limited section of the reservoir, reducing the representativeness of the results.

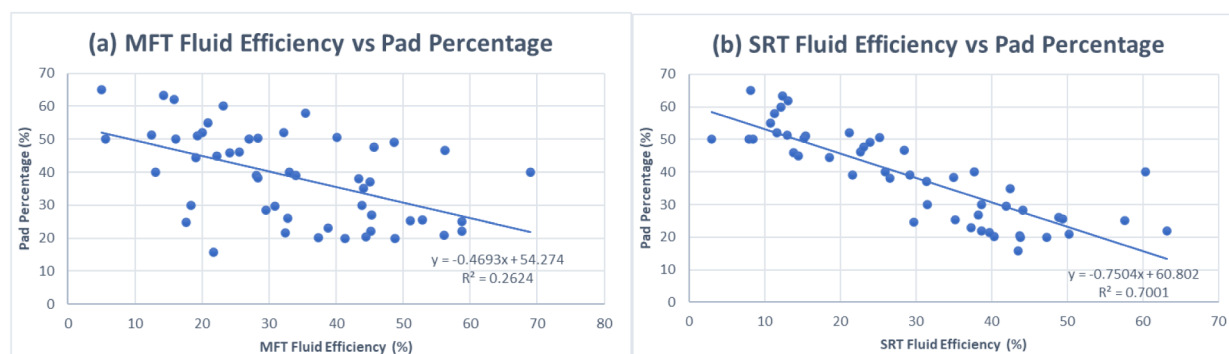


Figure 3—Linear correlation between fluid efficiency of tests and pad percentage (a) Mini-Falloff Tests (b) Step-Rate Tests

As a result, brine fluid efficiency derived from Step-Rate Tests was selected as a key predictive parameter for main fracture design, particularly for determining the optimal pad percentage. This approach was expected to reduce chemical usage and shorten operational time by eliminating the calibration injection tests, time-intensive and chemical-consuming phases.

Proposed Approaches

Reservoirs within the same geological compartment typically exhibit similar characteristics due to shared depositional environments, structural settings, and diagenetic histories. These compartments often possess comparable properties such as porosity, permeability, and stress orientation, which directly influence hydraulic fracturing operations. As a result, treatment responses, such as fluid efficiency, fracture geometry, and pressure behavior tend to follow similar trends within each compartment. This geological consistency

enables more reliable predictive modeling, as parameters derived from one well or stage within a compartment can be effectively applied to others in the same zone.

Following the identification of a strong correlation between fluid efficiency and pad percentage in **Section 2.1**, the N Area, which contains the largest number of data points, was selected for predictive model application. To enhance resolution and accuracy, the area was further subdivided into three geological compartments: C1, C2, and C3, based on subsurface characteristics and faults defined by the geologist team, as shown in **Figure 4** (left). The available fracturing stages within each compartment were distributed as follows: 4 stages in C1, 30 stages in C2, and 22 stages in C3, providing a dataset for compartment-specific model calibration and analysis (**Figure 4** (right)).

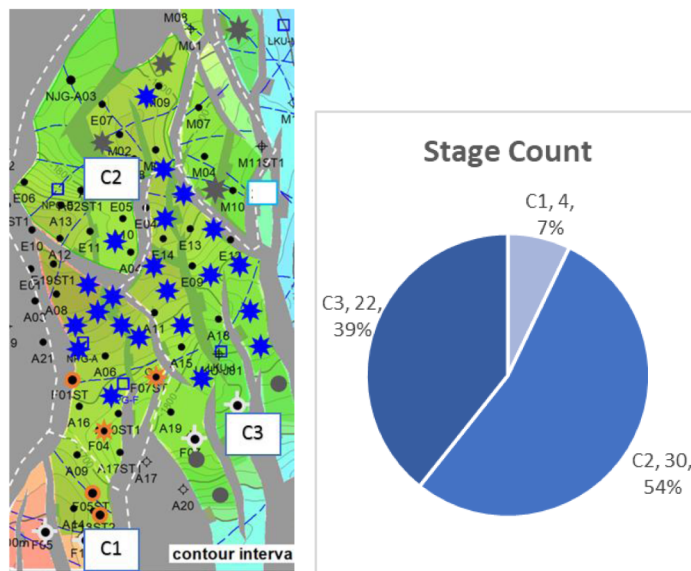


Figure 4—(Left) The N Area geological compartments: C1, C2, and C3, based on subsurface characteristics and faults (Right) The N Area fracturing stages count in each compartments

After grouping the dataset according to the geological compartments within the N Area, the analysis revealed stronger correlations within individual compartments compared to the combined dataset (**Figure 5**). By isolating data based on geological characteristics, the empirical linear relationships between Step-Rate Test fluid efficiency and pad percentage became more pronounced. The resulting R^2 values were 0.79, 0.75, and 0.79 for compartments C1, C2, and C3, respectively, each showing an improvement over the overall correlation of R^2 of 0.70 derived from all S-sands data in the N Area.

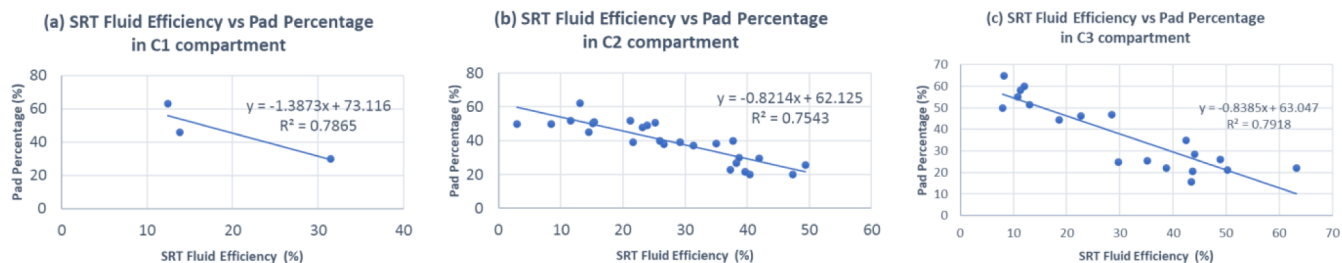


Figure 5—Linear correlation between Step-Rate Test fluid efficiency and pad percentage: (a) Compartment C1, (b) Compartment C2, (c) Compartment C3

With strong and consistent correlations established between Step-Rate Test fluid efficiency and pad percentage across all geological compartments in the N Area, the predictive model is considered ready for field application. The robustness of the model, supported by high R^2 values in each compartment,

demonstrates its reliability in forecasting treatment parameters based on test data in Step-Rate Test, omitting the Calibration Injection step, and utilizing data models from offset wells within the same geological compartment

Execution

To ensure the model's reliability prior to full-scale implementation, a field validation phase was conducted through trial stages. These trials aimed to assess the model's predictive accuracy under real operational conditions while applying the revised workflow, specifically, the omission of the calibration injection step. The validation was performed within the same well, Well-A*, across two stages. Stage 1 followed the conventional approach, including full treatment execution, allowing for direct comparison between actual test results and model predictions. Based on the outcome of Stage 1, Stage 2 proceeded with the revised method, omitting the calibration injection and relying solely on the predictive model developed from model within the same geological compartment. The results were then analyzed to evaluate the model's effectiveness and consistency.

*"Well-A" are anonymized designations used to maintain data confidentiality.

The first trial stages were conducted in Well-A to validate the predictive model under field conditions. In Stage 1 with conventional approach and full treatment execution, the model demonstrated a strong correlation with actual treatment results, showing only a 2% difference between the predicted pad percentage, from the SRT fluid efficiency and actual pad percentage with the model correlation, at 63.5% (Figure 6 (left)) and the actual outcome at 61.9% (Figure 6 right). This close agreement highlights the model's accuracy and practical applicability in real operations, reinforcing its potential for model usage in future stages.

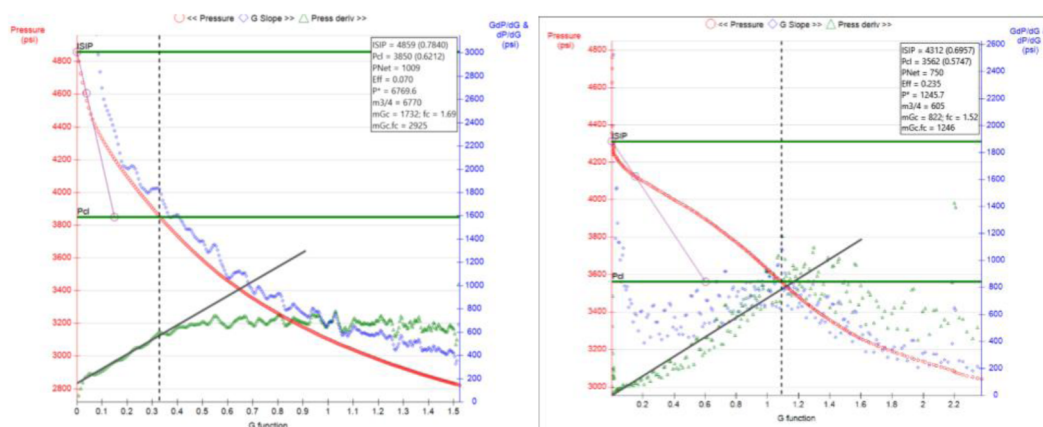


Figure 6—G-function plots from treatment execution in Well-A Stage 1: (Left) Step-Rate Test, (Right) Calibration Injection

In Well-A Stage 2, following the successful validation of the predictive model in the previous stage, the team proceeded by omitting the calibration injection step, as the operation was within the same geological compartment (S-sand). Utilizing the model predictions, the main hydraulic fracturing treatment was designed based on fluid efficiency obtained from the G-function plot of the Step-Rate Test (Figure 7 left). This approach resulted in a calculated pad volume of 49.8%. The main frac was successfully pumped according to the designed parameters, with actual treatment results closely matching the predicted values (Figure 7 right and Table 1). This outcome further validated the model's reliability and demonstrated its practical effectiveness for routine operations.

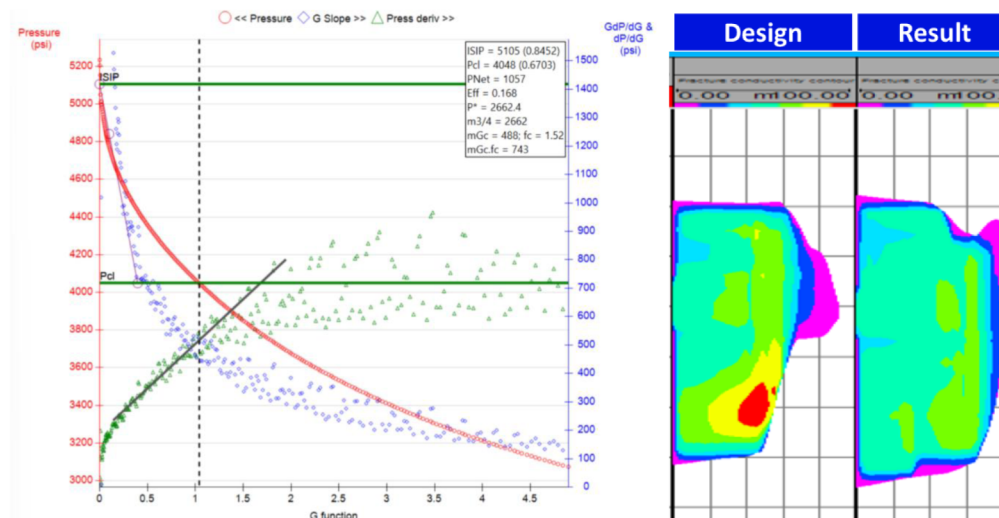


Figure 7—(Left) G-function plot from the Step-Rate Test in Well-A Stage 2, (Right) Fracture geometry and conductivity distribution from Well-A Stage 2

Table 1—Well A stage 2 fracture geometry values from design and actual results matched from pressure responses

	Propped Fracture Half-Length (m)	Propped Fracture Width (in)	Propped Fracture Height (m)	Effective Conductivity (md·ft)	Effective Fcd
Design	90.4	0.25	52.68	9993.98	18.02
Actual (from pressure matching)	88.67	0.23	58.31	8831.64	16.23

Following the successful outcomes demonstrated during the trial stages, the validated predictive modeling approach was implemented on a broader scale throughout 2023 and 2024. Specifically, these optimized practices were applied across 15 wells, encompassing a total of 26 fracturing stages within the N Area. This expansion significantly streamlined operational processes, enhancing both efficiency and economics in hydraulic fracturing projects.

Results Discussion

Throughout the application of the predictive model across all selected wells and stages, there were no recorded cases of failure or screen-out events, where pumping operations are typically halted due to insufficient pad volume or premature proppant bridging. This outcome highlights the model's effectiveness in accurately predicting treatment parameters, particularly pad percentage, ensuring sufficient fluid volume to maintain proppant transport and prevent operational disruptions. The consistent success across all stages reinforces the reliability of the model and its suitability for routine implementation in similar reservoir conditions.

The implementation of the predictive modeling approach resulted in significant operational and economic benefits. On average, 5 to 10 hours of operation time were saved per stage, representing a time reduction of approximately 20% compared to conventional workflows. This efficiency gain, primarily due to the omission of the calibration injection step and optimized fluid and chemical usage, contributed to a 7.6% average reduction in fracturing costs. Additionally, comparison between the design fracture geometry and the pressure-matched fracture geometry revealed minimal deviation, with calculated skin differences ranging from -10% to +7% (Figure 8). These results indicate that the model not only enhances operational efficiency but also maintains a high level of accuracy in treatment execution.

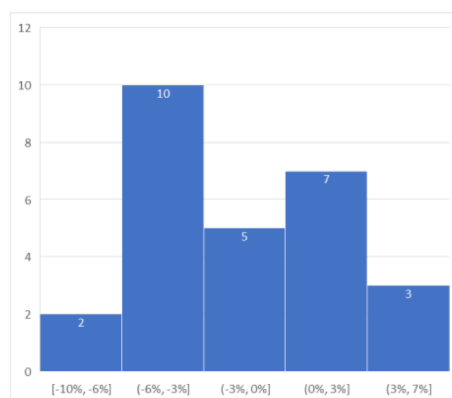


Figure 8—Calculated skin differences between the design fracture geometry and the pressure-matched fracture geometry

Future Potential Application

Building upon the robust empirical modeling demonstrated in this study, future developments could focus on integrating machine learning (ML) techniques to further enhance the predictive accuracy, generalizability, and scalability of hydraulic fracturing design, particularly in complex or data-limited environments. Recent research has underscored the transformative potential of ML in improving treatment placement accuracy, fluid efficiency prediction, and overall stimulation effectiveness.

For instance, [Almutary et al. \(2024\)](#) developed a supervised ML model trained on over 100 fracturing stages, which utilized a custom Placement Quality Index (PQI) derived from multidomain inputs including well logs, perforation design, and diagnostic pressure data. Their three-phase ML framework—comprising classification, regression, and optimization algorithms—achieved a prediction accuracy of 94% for stage success and significantly improved treatment outcomes when deployed in real-world field scenarios(SPE-222306-MS).

Moreover, [Khan et al. \(2024, 2025\)](#) introduced a physics-informed machine learning (PIML) and transfer learning approach to overcome limitations posed by sparse or incomplete datasets. Their synthetic physics-based datasets were used to pre-train neural networks, which were subsequently fine-tuned using real field data. These methods showed predictive improvements of up to 22% in key metrics like RMSE and MAPE, while integrating Particle Swarm Optimization (PSO) to optimize stimulation design for maximum productivity index(SPE-222903-MS; ITPC-24850-MS).

In addition to those efforts, [Akbari et al. \(2025\)](#) demonstrated the viability of using ML models to evaluate hydraulic fracturing performance with high accuracy using a range of operational, geological, and fracturing parameters. Their model achieved an average accuracy of 92% across multiple basins, showing that when sufficient structured data is available, supervised learning can reliably predict post-fracture production and assess the success of stimulation treatments. This reinforces the potential for ML to scale across diverse geological settings, particularly when paired with robust historical databases.

These academic advances suggest clear and promising directions for further deployment in the Sirikit Oil Field. By embedding ML and transfer learning into the current empirical framework, it is feasible to create generalized predictive models that reflect geological compartmentalization while remaining adaptable across other formations or basins. Incorporating domain-specific constraints, historical diagnostic test results, and fracture geometry feedback into the learning cycle could also improve model resilience, particularly in compartments with minimal prior calibration. Over time, this progression could lead to a unified, data-driven fracturing design system, streamlining workflows, minimizing dependency on calibration injections, and enabling faster and more cost-effective decision-making in both greenfield and brownfield developments.

Conclusion

This study introduces a predictive modeling framework for hydraulic fracturing design, developed and field-validated in the Sirikit Oil Field, with the aim of enhancing operational efficiency and minimizing reliance on conventional diagnostic steps. By analyzing over 200 fracturing stages and establishing a robust empirical correlation between fluid efficiency and pad percentage—particularly from Step-Rate Tests (SRTs)—the model demonstrated strong predictive capabilities. The incorporation of geological compartmentalization further improved accuracy, allowing treatment parameters to be reliably forecasted within consistent reservoir units. Field validation through trial stages in Well-A confirmed the model's practical utility, with a deviation of less than 2% between predicted and actual pad requirements.

Following validation, the model was successfully implemented across 15 wells and 26 stages in the N Area during 2023–2024. This deployment eliminated the need for Calibration Injection steps, with no recorded operational failures or screenouts. The approach yielded tangible operational benefits, including an average 20% reduction in stage execution time and a 7.6% cost saving per treatment. Moreover, the close agreement between design-based and pressure-matched fracture geometries—with minimal variation in skin values—affirmed the model's reliability and its adaptability across similar geological settings.

Looking ahead, the predictive framework established in this work offers a strong foundation for further development. Incorporating machine learning, especially physics-informed neural networks (PINNs) and transfer learning, presents a compelling avenue for enhancing model generalization, particularly in geologically complex or data-sparse areas. By embedding empirical correlations into a scalable, intelligent architecture, future models can support real-time decision-making, accelerate fracturing design processes, and expand usability across diverse formations beyond the Sirikit Oil Field. This would support the development of a universal, intelligent fracturing design system, offering both operational agility and data-driven reliability across broader unconventional asset portfolios.

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